

SAML

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Introduction to SAML in EU Login

EU Login offers support for the [Security Assertion Markup Language](#) (SAML) protocol.

SAML is a standardised means for conveying authentication data from a system capable of authenticating a user (an identity provider, "IdP") and a system that consumes user authentication data (a service provider, "SP").

SAML defines both message protocols i.e. the flows between the systems as well as the message contents.

The SAML message format is based on XML. It is enriched with necessary security data such as a digital signature of the message content.

The blocks which group information about an authenticated user and which are passed to the service provider are called *assertions*. Each assertion contains a number of *attributes*, which are the actual bits of user information.

The examples section below shows a [sample SAML response](#), such as it is currently provided by EU Login.

Currently, EU Login interacts with several service provider partners in SAML. One of those service providers is the *Juniper VPN* - commonly known as the [myremote portal](#).

Supported versions and profiles

EU Login supports the SAML versions **1.1** and **2.0**.

The currently implemented profiles are:

SAML 1.1	SAML 2.0
Browser/POST	Web Browser SSO

How to register a SAML Service Provider in EU Login

Environment

First of all we invite you to configure your Acceptance application with EU Login Acceptance.

The public URL of EU Login acceptance is <https://ecas.acceptance.ec.europa.eu/cas/login>.

The SAML metadata file for EU Login acceptance is accessible via HTTP:

- from outside the European Commission network: <https://ecas.acceptance.ec.europa.eu/cas/saml/metadata/ecas-acceptance-ec-europa-eu.xml>
- from inside the EC network: https://ecasa.cc.cec.eu.int:7002/cas/saml/metadata/ecasa-cc-cec-eu-int_7002.xml

Required information

EU Login is not a standard SAML product so it is not able to upload a SAML metadata file and magically work with your service provider.

Therefore, to use EU Login as an Identity Provider you have to provide the following information to EC-IAM-SERVICE-DESK@ec.europa.eu:

1. The **issuer URI** (aka entity ID) of your application. It has to correspond to the value of the `<saml2p:issuer/>` node in the SAML request received by EU Login.
2. The **assertion consumer URL**. The registered assertion consumer URL is a fallback if no consumer URL is defined in the SAML request received by EU Login.
3. One or more **authentication strenght(s)** accepted by your application. Strengths currently compatible with SAML are PASSWORD, PASSWORD_SMS, PASSWORD_TOKEN, PASSWORD_MOBILE_APP, MOBILE_APP, PASSWORD_SOFTWARE_TOKEN and STORK.
4. If you want activate the **strength upgrade** functionality (only possible if at least one multi-factor strength and at least one single-factor strength are accepted). Strength upgrade forces 2-factors authentication for users outside the trusted networks of the EU Institutions.
5. The **attributes** you want to receive in the SAML response and the **mapping** with supported [user details](#)
6. If you want to force authentication (**renew mode**), we recommend to use the attribute ForceAuthn="1" in the SAML Request. However, we also have a fallback on server side if your SAML client doesn't support that attribute.
7. Although the EU Login IdP accepts signed SAML requests as well as unsigned SAML requests, we strongly recommend to use **digital signature** in order to ensure the integrity of your SAML requests.
If you want to sign the SAML request, we need an **X.509 certificate containing the public key** to verify the digital signature.

Information required by EU Login	Corresponding entry in EU Login SAML 2.0 metadata file
Issuer URI	<code><md:EntityDescriptor ID="_683bc631-4bab-44b4-98d3-73667b03f128" entityID="https://ecasa.cc.cec.eu.int:7002/cas/login"></code>
Assertion consumer URL	<code><md:SPSSODescriptor ...></code> ... <code><md:AssertionConsumerService Binding="urn:oasis:names:tc:SAML:2.0:bindings:HTTP-POST" Location="https://ecasa.cc.cec.eu.int:7002/cas/login" index="0" isDefault="true" xsi:type="md:IndexedEndpointType"/></code> <code></md:SPSSODescriptor></code>
X.509 Certificate containing the public key used to verify signatures	<code><md:SPSSODescriptor ...></code> <code><md:KeyDescriptor use="signing"></code> <code><ds:KeyInfo></code> <code><ds:X509Data></code> ... <code><ds:X509Certificate></code> <i>MIIGXjCCBEagAwIBAgIQakKwI057uIV</i> <i>/ACZihKf91zANBgqhkiG9w0BAQ0FADCBzzELMAkGA1UE</i> <i>ThisIsThePublicKey</i> <i>wac8ukKIPUJfVGYOliPm3zyv4jxk1BH9+YmaA1eA1IIRZGH1w+</i> <i>mKVZ5EJfQ/1rBZEWhk17QFOXtu</i> <i>iX9eYGdGf3xQvVwKYJOBfSZTWtAHQlofsI3swKXiBBITnU5ayk=</i> <code></ds:X509Certificate></code> <code></ds:X509Data></code> <code></ds:KeyInfo></code> <code></md:KeyDescriptor></code> <code></md:SPSSODescriptor></code>

Table 1: EU Login metadata file as an example of how to find useful information to act as a SAML service provider

Supported User Details

The user information that EU Login can put in the SAML response is defined in the authenticationSuccessType in `ecas.xsd`, available at <https://ecas.ec.europa.eu/cas/schemas/ecas.xsd>.

For an example of how this information is returned, in the [sample SAML response](#) below take a look at the contents of the Assertion element.

SAML 2.0 message examples

Sample unsigned SAML request

The following shows a sample unsigned SAML 2.0 request; communication flow: Service Provider -> EU Login.

```
<samlp:AuthnRequest xmlns:samlp="urn:oasis:names:tc:SAML:2.0:protocol"
  Destination="https://ecasd.cc.cec.eu.int:7002/cas
/login"
  ForceAuthn="false"
  ID="_0xe010f8d9875b05b4f95a7806210b0be2"
  IsPassive="false"
  IssueInstant="2014-12-09T11:30:41.732Z"
  Version="2.0"
>
  <saml:Issuer xmlns:saml="urn:oasis:names:tc:SAML:2.0:assertion"
>d02di1022041dit</saml:Issuer>
</samlp:AuthnRequest>
```

Sample signed SAML request

The following shows a sample signed SAML 2.0 request; communication flow: Service Provider -> EU Login.

```

<samlp:AuthnRequest xmlns:samlp="urn:oasis:names:tc:SAML:2.0:protocol"
    Destination="https://ecasd.cc.cec.eu.int:7002/cas
/login"
    ForceAuthn="false"
    ID="_0x14956c887e664bdb71d7685b89b70619"
    IsPassive="false"
    IssueInstant="2014-12-09T11:38:29.418Z"
    Version="2.0"
>
    <saml:Issuer xmlns:saml="urn:oasis:names:tc:SAML:2.0:assertion"
>d02di1022041dit</saml:Issuer>
    <ds:Signature xmlns:ds="http://www.w3.org/2000/09/xmldsig#">
        <ds:SignedInfo>
            <ds:CanonicalizationMethod Algorithm="http://www.w3.org/2001/10
/xml-exc-c14n#" />
            <ds:SignatureMethod Algorithm="http://www.w3.org/2000/09
/xmldsig#rsa-sha1" />
            <ds:Reference URI="#_0x14956c887e664bdb71d7685b89b70619">
                <ds:Transforms>
                    <ds:Transform Algorithm="http://www.w3.org/2000/09
/xmldsig#enveloped-signature" />
                    <ds:Transform Algorithm="http://www.w3.org/2001/10/xml-
exc-c14n#WithComments">
                        <ec:InclusiveNamespaces xmlns:ec="http://www.w3.org
/2001/10/xml-exc-c14n#"
                            PrefixList="ds saml samlp"
                        />
                    </ds:Transform>
                </ds:Transforms>
                <ds:DigestMethod Algorithm="http://www.w3.org/2001/04
/xmlenc#sha256" />
                <ds:
DigestValue>GHmWZ3x+Ba+2YQS+BRx70st4DiuYlKZ4C6ZeidlST8Y=</ds:DigestValue>
                </ds:Reference>
            </ds:SignedInfo>
            <ds:SignatureValue>
XYDBDIb4ft1uqSBcEtWQ9UdELbjfz0bkIhzEA2pB2AEqvbZLp3bI7vrIzesImIXqL8iWtp5VBAq
W
UsWvM0tXI+wliNC013rUNiKM9q4Z5lF0lyAcrVH3RpSfrkK0VxYHAVwk3ipewFHPJcack6UuEpI
7
YDYz+vsnf1xXK
/nwY8ABzNb4ZPkPvKagIhP86Cu8iRf+kHgawLJvX7kqQ8Dd4dlnFDKpE+3ypiu
ocCw5Yy/jhzG/uEYny
/cSYjntdfMb9VhxuIMIGh9kGP17DARwGEct9yevdlQXxkC0q18fKs+FEra
0E14dlpm5/OGmlaPv219DHCsJmmySrD1IyfyIw==
</ds:SignatureValue>
        </ds:Signature>
    </samlp:AuthnRequest>

```

Sample SAML request with the AssertionConsumerServiceURL attribute

The following shows a sample unsigned SAML 2.0 request containing the AssertionConsumerServiceURL attribute ; communication flow: Service Provider -> EU Login.

```
<samlp:AuthnRequest xmlns:samlp="urn:oasis:names:tc:SAML:2.0:protocol"
    AssertionConsumerServiceURL="https://webgate.
d02di1022041dit.ec.europa.eu/saml/sso"
    Destination="https://ecasd.cc.cec.
eu.int:7002/cas/login"
    ForceAuthn="false"
    ID="_0xe010f8d9875b05b4f95a7806210b0be2"
    IsPassive="false"
    IssueInstant="2014-12-09T11:30:41.732Z"
    Version="2.0"
>
    <saml:Issuer xmlns:saml="urn:oasis:names:tc:SAML:2.0:assertion"
>d02di1022041dit</saml:Issuer>
</samlp:AuthnRequest>
```

EU Login uses the value of the SAML attribute as the destination of the SAML response (i.e. <https://webgate.d02di1022041dit.ec.europa.eu/saml/sso>) instead of the value registered in EU Login.

Sample SAML response

The following is a sample SAML 2.0 response; communication flow: EU Login -> Service Provider.

```
<saml2p:Response xmlns:saml2p="urn:oasis:names:tc:SAML:2.0:protocol" xmlns:
saml2="urn:oasis:names:tc:SAML:2.0:assertion" xmlns:xs="http://www.w3.org
/2001/XMLSchema" Consent="urn:oasis:names:tc:SAML:2.0:consent:obtained"
Destination="https://myremote.ec.europa.eu/dana-na/auth/saml-consumer.cgi"
ID="_74dba72f-47ba-4ab5-94a2-7fe5abc56768" InResponseTo="
_76358f4621f1944be270f68e9b8e9300" IssueInstant="2012-10-23T06:42:30.267Z"
Version="2.0">
    <saml2:Issuer Format="urn:oasis:names:tc:SAML:2.0:nameid-format:entity"
>https://ecas.cc.cec.eu.int:7002/cas/login</saml2:Issuer>
    <ds:Signature xmlns:ds="http://www.w3.org/2000/09/xmldsig#">
        <ds:SignedInfo>
            <ds:CanonicalizationMethod Algorithm="http://www.w3.org/2001/10
/xml-exc-c14n#" />
            <ds:SignatureMethod Algorithm="http://www.w3.org/2001/04
/xmldsig-more#rsa-sha256" />
            <ds:Reference URI="#_74dba72f-47ba-4ab5-94a2-7fe5abc56768">
                <ds:Transforms>
                    <ds:Transform Algorithm="http://www.w3.org/2000/09
/xmldsig#enveloped-signature" />
                    <ds:Transform Algorithm="http://www.w3.org/2001/10/xml-
exc-c14n#" />
                    <ec:InclusiveNamespaces xmlns:ec="http://www.w3.org
/2001/10/xml-exc-c14n#" PrefixList="xs" />
                </ds:Transform>
            </ds:Transforms>
            <ds:DigestMethod Algorithm="http://www.w3.org/2001/04
/xmldsig#sha256" />
```

```
<ds:DigestValue>JyrRua
/49aFjNXPJoYWFsXZhBbJwevCYJowNhnTmMvo=</ds:DigestValue>
</ds:Reference>
</ds:SignedInfo>
<ds:
SignatureValue>aP1+3Tey9leMqAub8oEC0aWzG2CG0GGHpG0WYhE2fB838PCdyo3GZrN0Hmn
0PLZy9p8Y86VL79XMETiu5KnvDaKGtw1IN19BnYqyRgytDjb/sqfzHl124EpyYSUGeLFi
/dILUjP9wBm7k1rjrDp7uVj6ULBo1iRvFS+iGEmFKPMm1UHMMW7KqKteUtcVbA1pma
/xffm0XxlyZcFhe5dmPMiPJ1DtbfxjvW3snbCoR2NhfMwIbPkQCL4B2oNoSe0zAk4WDpfUsxB16
J9MHxLK90Au6ybBswsbZoYezE6TWlhI5YY0UUKAC7+pRIOLxSRu9Yan1uhZ1jx7ecP4HR5Qw==<
/ds:SignatureValue>
<ds:KeyInfo>
<ds:X509Data>
<ds:X509SubjectName>CN=ecas-prod-sts,1.2.840.113549.1.9.1
=#162344494749542d454341532d444556454c4f504d454e544065632e6575726f70612e657
5,0U=DIGIT,0=European Commission,L=Brussels,C=BE</ds:X509SubjectName>
<ds:
X509Certificate>MIIGeTCCBGgAwIBAgIQVtSzs2ZsuMRXGcttBnBlfjANBgkqhkiG9w0BAQU
FADCB5DELMAGGA1UE
...
Qy0Gt8gWsjs=</ds:X509Certificate>
<ds:
X509Certificate>MIIHXjCCBUagAwIBAgIBATANBgkqhkiG9w0BAQUFADCB5DELMAGGA1UEBHM
CQkUxETAPBgNVBAGM
...
Qlq5pIka4rlm</ds:X509Certificate>
</ds:X509Data>
</ds:KeyInfo>
</ds:Signature>
<saml2p:Status>
<saml2p:StatusCode Value="urn:oasis:names:tc:SAML:2.0:status:
Success"/>
<saml2p:StatusMessage>Successful EU Login authentication</saml2p:
StatusMessage>
</saml2p:Status>
<saml2:Assertion ID="_79ba7d43-dc55-430d-8c55-ff363b735bf2"
IssueInstant="2012-10-23T06:42:30.267Z" Version="2.0">
<saml2:Issuer Format="urn:oasis:names:tc:SAML:2.0:nameid-format:
entity">https://ecas.cc.cec.eu.int:7002/cas/login</saml2:Issuer>
<saml2:Subject>
<saml2:NameID Format="urn:oasis:names:tc:SAML:1.1:nameid-
format:unspecified">keschma</saml2:NameID>
<saml2:SubjectConfirmation Method="urn:oasis:names:tc:SAML:2.0:
cm:bearer">
<saml2:SubjectConfirmationData InResponseTo="
_76358f4621f1944be270f68e9b8e9300" NotOnOrAfter="2012-10-23T07:12:30.267Z"
Recipient="https://myremote.ec.europa.eu/dana-na/auth/saml-consumer.cgi"/>
</saml2:SubjectConfirmation>
</saml2:Subject>
<saml2:Conditions NotBefore="2012-10-23T06:12:30.267Z"
NotOnOrAfter="2012-10-23T07:12:30.267Z">
<saml2:AudienceRestriction>
<saml2:Audience>https://myremote.ec.europa.eu/dana-na/auth
```

```

/saml-endpoint.cgi?p=sp1</saml2:Audience>
  </saml2:AudienceRestriction>
    <saml2:OneTimeUse/>
    <saml2:ProxyRestriction Count="0"/>
  </saml2:Conditions>
  <saml2:AuthnStatement AuthnInstant="2012-10-23T06:42:30.267Z"
SessionIndex="_d4a2eb81-4ee9-45f9-b0c8-cfd252317c0e" SessionNotOnOrAfter="
2012-10-23T07:12:30.267Z">
    <saml2:AuthnContext>
      <saml2:AuthnContextClassRef>urn:oasis:names:tc:SAML:2.0:ac:
classes:Password</saml2:AuthnContextClassRef>
    </saml2:AuthnContext>
  </saml2:AuthnStatement>
  <saml2:AttributeStatement>
    <saml2:Attribute FriendlyName="User Identifier" Name="user"
NameFormat="urn:oasis:names:tc:SAML:2.0:attrname-format:basic">
      <saml2:AttributeValue xmlns:xsi="http://www.w3.org/2001
/XMLSchema-instance" xsi:type="xs:string">keschma</saml2:AttributeValue>
    </saml2:Attribute>
    <saml2:Attribute FriendlyName="Email" Name="email" NameFormat="
urn:oasis:names:tc:SAML:2.0:attrname-format:basic">
      <saml2:AttributeValue xmlns:xsi="http://www.w3.org/2001
/XMLSchema-instance" xsi:type="xs:string">Martin.HOFFMANN@ec.europa.eu<
/saml2:AttributeValue>
    </saml2:Attribute>
    <saml2:Attribute FriendlyName="Organisation" Name="domain"
NameFormat="urn:oasis:names:tc:SAML:2.0:attrname-format:basic">
      <saml2:AttributeValue xmlns:xsi="http://www.w3.org/2001
/XMLSchema-instance" xsi:type="xs:string">eu.europa.ec</saml2:
AttributeValue>
    </saml2:Attribute>
    <saml2:Attribute FriendlyName="Organisation Username" Name="
domainUsername" NameFormat="urn:oasis:names:tc:SAML:2.0:attrname-format:
basic">
      <saml2:AttributeValue xmlns:xsi="http://www.w3.org/2001
/XMLSchema-instance" xsi:type="xs:string">keschma</saml2:AttributeValue>
    </saml2:Attribute>
    <saml2:Attribute FriendlyName="First Name" Name="firstName"
NameFormat="urn:oasis:names:tc:SAML:2.0:attrname-format:basic">
      <saml2:AttributeValue xmlns:xsi="http://www.w3.org/2001
/XMLSchema-instance" xsi:type="xs:string">Martin</saml2:AttributeValue>
    </saml2:Attribute>
    <saml2:Attribute FriendlyName="Last Name" Name="lastName"
NameFormat="urn:oasis:names:tc:SAML:2.0:attrname-format:basic">
      <saml2:AttributeValue xmlns:xsi="http://www.w3.org/2001
/XMLSchema-instance" xsi:type="xs:string">HOFFMANN</saml2:AttributeValue>
    </saml2:Attribute>
    <saml2:Attribute FriendlyName="Groups" Name="groups"
NameFormat="urn:oasis:names:tc:SAML:2.0:attrname-format:basic">
      <saml2:AttributeValue xmlns:xsi="http://www.w3.org/2001
/XMLSchema-instance" xsi:type="xs:string">DIGIT_WEB_ALL</saml2:
AttributeValue>
    <saml2:AttributeValue xmlns:xsi="http://www.w3.org/2001

```



```
/XMLSchema-instance" xsi:type="xs:string">INTERNET</saml2:AttributeValue>
    <saml2:AttributeValue xmlns:xsi="http://www.w3.org/2001
/XMLSchema-instance" xsi:type="xs:string">LIVENEWS</saml2:AttributeValue>
    <saml2:AttributeValue xmlns:xsi="http://www.w3.org/2001
/XMLSchema-instance" xsi:type="xs:string">DG_DIGIT</saml2:AttributeValue>
    <saml2:AttributeValue xmlns:xsi="http://www.w3.org/2001
/XMLSchema-instance" xsi:type="xs:string">SDT_VISTA</saml2:AttributeValue>
    <saml2:AttributeValue xmlns:xsi="http://www.w3.org/2001
/XMLSchema-instance" xsi:type="xs:string">EUROPOL</saml2:AttributeValue>
    <saml2:AttributeValue xmlns:xsi="http://www.w3.org/2001
/XMLSchema-instance" xsi:type="xs:string">SG_CISNET</saml2:AttributeValue>
    <saml2:AttributeValue xmlns:xsi="http://www.w3.org/2001
/XMLSchema-instance" xsi:type="xs:string">PRICEFIX</saml2:AttributeValue>
    <saml2:AttributeValue xmlns:xsi="http://www.w3.org/2001
/XMLSchema-instance" xsi:type="xs:string">DIGIT_ITBOS</saml2:
AttributeValue>
    <saml2:AttributeValue xmlns:xsi="http://www.w3.org/2001
/XMLSchema-instance" xsi:type="xs:string">DIGIT_VLE_P</saml2:
AttributeValue>
    <saml2:AttributeValue xmlns:xsi="http://www.w3.org/2001
/XMLSchema-instance" xsi:type="xs:string">OXAN</saml2:AttributeValue>
    <saml2:AttributeValue xmlns:xsi="http://www.w3.org/2001
/XMLSchema-instance" xsi:type="xs:string">DIGIT_ECAS_A</saml2:
AttributeValue>
    <saml2:AttributeValue xmlns:xsi="http://www.w3.org/2001
/XMLSchema-instance" xsi:type="xs:string">ARES_USERS</saml2:AttributeValue>
    <saml2:AttributeValue xmlns:xsi="http://www.w3.org/2001
/XMLSchema-instance" xsi:type="xs:string">DEV_IQSG</saml2:AttributeValue>
    <saml2:AttributeValue xmlns:xsi="http://www.w3.org/2001
/XMLSchema-instance" xsi:type="xs:string">OPOCE_LEXAADM</saml2:
AttributeValue>
    <saml2:AttributeValue xmlns:xsi="http://www.w3.org/2001
/XMLSchema-instance" xsi:type="xs:string">OIB_KW_PARENT</saml2:
AttributeValue>
    <saml2:AttributeValue xmlns:xsi="http://www.w3.org/2001
/XMLSchema-instance" xsi:type="xs:string">RAPIDW</saml2:AttributeValue>
    <saml2:AttributeValue xmlns:xsi="http://www.w3.org/2001
/XMLSchema-instance" xsi:type="xs:string">DANTE</saml2:AttributeValue>
    <saml2:AttributeValue xmlns:xsi="http://www.w3.org/2001
/XMLSchema-instance" xsi:type="xs:string">PRICEMOB</saml2:AttributeValue>
    <saml2:AttributeValue xmlns:xsi="http://www.w3.org/2001
/XMLSchema-instance" xsi:type="xs:string">CELEXW</saml2:AttributeValue>
    <saml2:AttributeValue xmlns:xsi="http://www.w3.org/2001
/XMLSchema-instance" xsi:type="xs:string">OIB_KIDDYWEB</saml2:
AttributeValue>
    <saml2:AttributeValue xmlns:xsi="http://www.w3.org/2001
/XMLSchema-instance" xsi:type="xs:string">RAEPLUS</saml2:AttributeValue>
    <saml2:AttributeValue xmlns:xsi="http://www.w3.org/2001
/XMLSchema-instance" xsi:type="xs:string">RELEX_TARIQAEC</saml2:
AttributeValue>
    <saml2:AttributeValue xmlns:xsi="http://www.w3.org/2001
/XMLSchema-instance" xsi:type="xs:string">SLWF</saml2:AttributeValue>
</saml2:Attribute>
```



```

    </saml2:AttributeStatement>
  </saml2:Assertion>
</saml2p:Response>

```

Troubleshooting

Error on EU Login page after redirection	Cause
"SAML AuthnRequest must contain an Issuer"	The SAML request doesn't contain any <saml:Issuer/> entry.
"The requested relying party is not registered (SAML). Please register your application with the IAM (Identity and Access Management) service"	The value within <saml:Issuer/> doesn't correspond to any issuer URI registered in EU Login.
"Invalid SAML AuthnRequest signature"	The verification of the digital signature failed. Check if the private key used to sign and the public key registered in EU Login are parts of the same valid key pair.
"Unsigned SAML AuthnRequest while EU Login is expecting a signed SAML AuthnRequest"	A public key is linked to the current issuer in EU Login but the SAML request is not signed.

Note: If EU Login receives a signed SAML request but no public key is linked to the issuer URI, the signature is not verified, the SAML request is accepted by EU Login and the authentication web flow continues.

Annex

Sequence Diagram

